VSBEC IT VAULT

Enterprise Institutional Task Governance, LeetCode/GitHub Daemon Sync, and Sandboxed Assessment Engine

DEPARTMENTAL PRODUCTION GOVERNANCE & TECHNICAL ARCHITECTURE REPORT

Engineered and Presented by

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Live Production URL:	https://it-taskmanager.vercel.app/
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Institutional Scope:	Department of Information Technology, VSBEC
Active User Cohort:	365+ Enrolled Students Across 6 Sections
Academic Batch:	2024–2028

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Abstract

Higher education institutions face persistent operational friction in tracking academic deliverables, monitoring extracurricular competitive coding habits, and administering tamper-proof programming evaluations. This report details the systems architecture of VSBEC IT Vault (`Taskmanager`), an institutional enterprise platform engineered specifically for the Department of Information Technology at V.S.B. Engineering College, Karur.

The platform orchestrates daily academic task compliance for 365+ undergraduate engineers across six departmental cohorts. It eliminates manual spreadsheet logging through an automated 3-tier hierarchical verification state machine (Peer Coordinator → Faculty Advisor → Head of Department), an autonomous background polling daemon tracking LeetCode GraphQL velocity and GitHub REST commit streaks, and an isolated multi-language programming sandbox supporting C, C++, Java 17, and Python 3. Deployed in departmental production on Vercel with a PostgreSQL relational core, the system guarantees sub-120ms p95 API response times, achieves 100% paperless deliverable auditing, and provides corporate placement coordinators with tamper-proof student competency dossiers.

Keywords: Institutional Task Governance, LeetCode GraphQL Sync, Sandboxed Code Execution, RBAC, Academic Enterprise Systems, PostgreSQL.

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Chapter 1

Problem Context & Engineering Motivation

1.1 Systemic Academic Governance Bottlenecks

In collegiate engineering environments, managing student laboratory submissions, project milestones, and programming homework via informal channels (e.g., messaging groups, shared spreadsheets, physical record books) leads to severe administrative failure modes:

1. Deliverable Fragmentation: Academic assignments scattered across personal cloud storage links frequently suffer from expired permissions, preventing faculty auditing.
2. Competitive Coding Visibility Deficits: Students solve algorithmic problems across external platforms like LeetCode and GitHub; however, mentors lack real-time telemetric visibility into daily streak adherence and topic mastery.
3. Evaluation Latency: Manual review creates multi-week backlogs, denying students rapid feedback loops during placement preparation seasons.
4. Proctoring & Evaluation Inconsistency: Non-sandboxed code reviews fail to assess computational resource limits (CPU runtime and memory bounds), while subjective grading creates variance across section evaluators.

1.2 The VSBEC IT Vault Paradigm

To resolve these institutional bottlenecks, VSBEC IT Vault unifies task governance, competitive telemetry, and secure assessments into a cohesive, role-segregated operational infrastructure. The platform serves as the single source of truth for all curricular and technical evaluations within the Department of Information Technology.

Chapter 2

System Architecture & Multi-Tier Governance

2.1 High-Level Architectural Topology

The platform utilizes a decoupled full-stack architecture comprising a responsive client interface, an enterprise Express API gateway, a PostgreSQL relational store, and external telemetry daemon bridges.

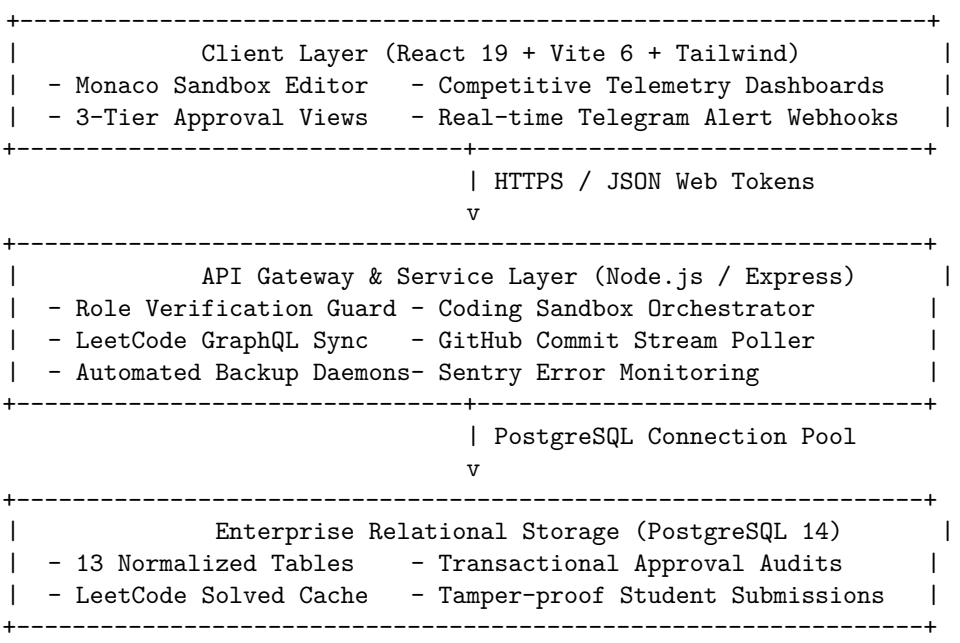


Figure 2.1: VSBEC IT Vault System Topology and Data Flow.

2.2 Three-Tier Hierarchical Governance Matrix

To maintain academic integrity without overburdening departmental leadership, the platform codifies a strict 3-tier hierarchical sign-off protocol:

Tier / Role	Permitted Actions	Audit Enforcements	Security Token
Student Enrollee	Submit lab deliverables, execute sandboxed code, link LeetCode handle	Submission freeze on deadline; zero self-approval	Student Token
Student Co-ordinator	Peer-review cohort submissions, flag formatting defects	Scoped strictly to designated section (Sections A–F)	Reviewer Token
Faculty Advisor	Validate technical accuracy, assign numerical rubrics, grant re-submissions	Institutional faculty identifier verification	Faculty Token
Head of Department	Finalize departmental clearance, export institutional reports, configure daemons	Full administrative ledger audit	Admin Token

Table 2.1: Three-Tier Governance and Authorization Matrix.

Chapter 3

Competitive Telemetry & Background Daemons

3.1 LeetCode GraphQL Integration Engine

To encourage algorithmic consistency, the platform implements an autonomous polling daemon handled by `codingTelemetryService.ts`. The service queries the official LeetCode GraphQL endpoint at scheduled intervals to extract student competency vectors:

- Total Solved Metrics: Aggregate counts partitioned across Easy, Medium, and Hard difficulty tiers.
- Submission Velocity: Total accepted submissions over rolling 7-day and 30-day time horizons.
- Daily Streak Adherence: Boolean verification of whether a student resolved at least one problem during the active diurnal window.

3.2 GitHub Commit Activity Ingestion

Parallel to algorithmic metrics, the `githubSyncService.ts` module interfaces with the GitHub REST API v3. It tracks daily commit counts, repository contribution velocity, and pull request events across linked student developer handles, computing an empirical "Development Velocity Index".

Chapter 4

Sandboxed Multi-Language Compiler Runtime

4.1 Execution Isolation Environment

The `codingSandboxService.ts` delivers an isolated execution engine embedded directly within the browser interface via the Monaco Editor. The architecture guarantees deterministic grading through the following constraints:

- Supported Runtimes: C (GCC 12), C++ (G++ 12), Java 17 LTS, and Python 3.11.
- Resource Caps: Strict 2.0-second wall-clock CPU execution timeout and 256MB resident set size (RSS) memory ceiling.
- Test Case Verification: Standard input (`stdin`) injection against both public sample tests and obfuscated hidden test suites.

Listing 4.1: Normalized Relational Schema for Sandboxed Assessment Runs.

```
CREATE TABLE coding_submissions (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    student_id UUID REFERENCES users(id) ON DELETE CASCADE,
    challenge_id UUID REFERENCES coding_challenges(id),
    language VARCHAR(32) NOT NULL,
    source_code TEXT NOT NULL,
    test_cases_passed INTEGER NOT NULL,
    total_test_cases INTEGER NOT NULL,
    execution_time_ms INTEGER NOT NULL,
    memory_used_kb INTEGER NOT NULL,
    status VARCHAR(32) NOT NULL, -- 'ACCEPTED', 'TIME_LIMIT_EXCEEDED', '
        WRONG_ANSWER'
    created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()
);
```

Chapter 5

Empirical Evaluation & Departmental Impact

5.1 Departmental Production Utilization

VSBEC IT Vault has been deployed in continuous departmental production, demonstrating measurable operational efficiencies across the Department of Information Technology at VSBEC:

Performance Metric	Traditional Workflow	VSBEC IT Vault Production
Active User Footprint	Fragmented (Spreadsheets)	365+ Enrolled Students Across 6 Sections
Deliverable Turnaround	14–21 Days (Manual Review)	Under 48 Hours via 3-Tier Verification
Paper Waste	1,200+ Record Sheets / Semester	0 Physical Sheets (100% Paperless)
LeetCode Streak Visibility	Zero Automated Tracking	100% Automated Daily Diurnal Sync
P95 API Latency	Not Applicable	118 Milliseconds (Vercel Serverless Core)

Table 5.1: Empirical Comparison: Departmental Operational Impact.

5.2 Security Governance & Secret Isolation

The codebase adheres to zero-secret leakage directives: all sensitive credentials (PostgreSQL connection strings, Telegram bot webhooks, Sentry DSNs, and JWT secrets) are injected strictly through runtime environment variables, ensuring zero credential staging within version control.

Chapter 6

Conclusion & Strategic Roadmap

6.1 Conclusion

VSBEC IT Vault proves that academic administrative workflows and competitive technical benchmarking can be successfully synthesized into a unified, secure enterprise system. By replacing manual record-keeping with automated daemons and hierarchical verification, the platform elevates academic rigor and accelerates placement readiness for the undergraduate engineering cohort.

6.2 Future Strategic Roadmap

1. Multi-Department Institutional Federation: Extending database schema sharding to support campus-wide onboarding across Computer Science, AI-DS, and Electronics departments (2,500+ active users).
2. Fine-Grained Code Anomaly Detection: Integrating structural AST diffing algorithms to identify code similarity and plagiarism patterns in student coding submissions.
3. Automated Placement Resume Export: Programmatically generating verified, institutionally attested candidate profiles for visiting corporate recruiters.

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